# Archita Srivastava

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## PROFESSIONAL SUMMARY

Robotics and ML student researcher passionate about perception and autonomy for physical robots.

## EDUCATION

**Simon Fraser University (SFU)** **Burnaby, BC**
*B.Sc. Computer Science, Specialization in AI/ML, Minor in Statistics* *Expected May 2027*

- **GPA:** 3.94 / 4.33
- **Courses**: Robotic Autonomy, Deep Learning, Machine Learning, Systems Programming, Quantum Algorithms
- **Leadership:** Technical Lead, Women in Computing Science (2023–24); Acting VP, Google Developer Group SFU (2025)
- **Awards & Scholarships:** DAAD Scholarship; T-CAIREM Research Grant; Generation Google Scholarship (North America); Peng Tu Ge Scholarship for Women in CS; Bit Quill Technologies Scholarship for Women in CS; WICS Alumni Award; Justin Xie Memorial Award

## EXPERIENCE

**Undergraduate Research Assistant** **Sept 2026 – Present**
*Introbotics Lab, Simon Fraser University* *Burnaby, BC*

- Engineering a pipeline that refines generative 3D object priors with real robot observations to produce physically grounded, articulated assets for manipulation

**Robotics Research Intern – DAAD RISE** **May 2026 – Sept 2026**
*MAIRO Lab, Karlsruhe Institute of Technology* *Karlsruhe, Germany*

- Developed the segmentation head of a Mixture-of-Experts (MoE) model for outdoor scene understanding, implementing early- and mid-fusion strategies to use RGB and LiDAR data for 2D semantic segmentation
- Implemented data loaders for large-scale multimodal datasets (Waymo Perception, KITTI-360)
- Integrated a PCL-based VDB map generator with an A* planner to convert segmentation output into navigable terrain maps for the ANYmal quadruped
- Built a test controller in ROS2 and validated the map generation, path planning, and controller execution pipeline in RViz
- Tested locomotion policies on a Unitree Go2 quadruped with an egocentric depth camera mounted onboard

**Undergraduate Research Assistant** **Jan 2026 – Jun 2026**
*Reliable Systems Lab, Simon Fraser University* *Burnaby, BC*

- Investigated the memory behavior of a quantized 70B-parameter LLM on llama.cpp under memory-mapped (mmap) versus non-mmap inference, informing deployment strategies for resource-constrained compute
- Evaluated on-device LLM inference (Llama 3.2 1B, Gemma 2 2B, SmolLM2 1.7B) on a Google Pixel 10 Pro using llama.cpp with Perfetto tracing to analyze memory behavior across varying context lengths

**AI Research Intern – T-CAIREM** **May 2025 – Aug 2025**
*Martel Lab, University of Toronto (Sunnybrook Research Institute)* *Toronto, ON*

- Curated a 5,400+ report training corpus for PHI de-identification, combining manually annotated Sunnybrook reports with synthetic and public clinical datasets
- Fine-tuned Llama 3.1 8B using LoRA (PEFT), improving F1 by 24% over a GLiNER baseline
- Compared medical VLMs for mammogram malignancy and density classification via linear probing and zero-shot inference

**Machine Learning Engineering Intern** **May 2024 – Dec 2024**
*Royal Bank of Canada* *Toronto, ON*

- Enriched Avion Rewards merchant data by combining enterprise Snowflake records with web-scraped sources in ChromaDB, increasing average data coverage by 87%
- Implemented a RAG-based retrieval system and a LightFM recommender to rank results for personalized semantic search

## PROJECTS

**Follow-Ahead Robot** **Jan 2026 – May 2026**
*Project Link*

- Led hardware integration for a 4-person team reproducing Leisiazar et al.'s follow-ahead framework (MCTS + A2C + LSTM), migrating the codebase from ROS1 to ROS2 and configuring a VICON motion-capture system to track robot and human poses
- Built the VICON-ROS2 bridge to deploy on a QBot 2e, physically validating across six trajectory types and achieving up to 84.7% follow-ahead success on slow trajectories

## SKILLS

**Robotics:** ROS2, IsaacSim, IsaacLab, Gazebo, RViz, VICON, ANYmal, Unitree Go2, QBot 2e
**Languages:** Python, C/C++, R, SQL
**AI/ML:** PyTorch, Hugging Face, Scikit-learn, NumPy, Pandas, LoRA, llama.cpp
**Tools:** Linux, Git, Docker, AWS, HPC SLURM, Perfetto

## PUBLICATIONS & PRESENTATIONS

- Presenter (DAAD RISE Conference 2026, Heidelberg, Germany): "More Than Meets the Eye: Multimodal Semantic Scene Understanding in Robotics"
- Winner, Best Presentation (T-CAIREM 2025): "VLMs for Pre-Surgical Prediction of Upstaging in DCIS"
- Co-Author (ACII 2023, MIT): Contextual Emotion Est. from Image Captions; awarded 1st Place at SFU Research Symposium